

ENERGY: sMAPE at each forecast step
Lookback = 24 steps, Forecast horizon = 72 steps

Forecast step	CNN	LSTM	dCeNN+ELM	dCeNN+ELM+ASP
1.000	56.359	68.664	68.275	46.456
2.000	58.165	63.632	69.031	47.253
3.000	59.906	62.103	70.039	48.264
4.000	61.076	62.215	71.060	49.297
5.000	61.856	62.897	71.487	49.732
6.000	63.445	63.615	71.817	50.068
7.000	64.503	64.298	72.545	50.805
8.000	64.896	65.195	73.265	51.519
9.000	65.893	65.856	73.515	51.751
10.000	66.332	66.405	73.323	51.563
11.000	66.625	66.786	73.062	51.298
12.000	67.337	67.147	73.499	51.745
13.000	67.451	67.441	74.229	52.471
14.000	68.014	67.673	74.886	53.126
15.000	68.266	67.825	75.073	53.315
16.000	68.647	67.940	74.689	52.831
17.000	69.054	68.014	74.368	52.365
18.000	69.421	68.112	74.095	52.016
19.000	69.085	68.232	73.777	51.752
20.000	69.260	68.355	74.163	52.140
21.000	69.363	68.506	75.128	53.249
22.000	69.384	68.572	76.699	54.923
23.000	69.383	68.672	77.819	56.065
24.000	69.512	68.927	78.348	56.594
25.000	69.877	69.255	79.005	57.247
26.000	70.140	69.460	79.828	58.063
27.000	70.359	69.578	80.143	58.375
28.000	70.175	69.648	79.573	57.809
29.000	70.116	69.710	79.049	57.279
30.000	70.583	69.821	79.165	57.403
31.000	70.557	69.926	79.510	57.793
32.000	70.317	70.021	80.018	58.297
33.000	70.707	70.126	79.907	58.187
34.000	71.041	70.212	79.922	58.185
35.000	70.911	70.224	79.901	58.160
36.000	70.802	70.255	80.644	58.899
37.000	70.797	70.283	82.513	60.761
38.000	71.048	70.321	84.200	62.440
39.000	71.047	70.318	84.860	63.100
40.000	71.142	70.311	85.032	63.260
41.000	71.330	70.304	85.274	63.385
42.000	71.166	70.301	84.812	62.847
43.000	71.067	70.309	84.158	62.011
44.000	71.189	70.333	84.144	61.823
45.000	71.375	70.327	84.974	62.651
46.000	71.491	70.341	85.415	63.225
47.000	71.281	70.358	84.433	62.587
48.000	71.159	70.516	83.603	61.842
49.000	71.136	70.719	83.083	61.321
50.000	71.388	70.798	83.067	61.301
51.000	71.372	70.822	82.877	61.109
52.000	71.408	70.822	81.825	60.057
53.000	71.251	70.811	81.131	59.369
54.000	71.377	70.809	80.848	59.092
55.000	71.276	70.874	81.957	60.200
56.000	71.056	70.934	83.761	62.004
57.000	71.297	70.995	85.723	63.973
58.000	71.697	71.018	85.589	63.837
59.000	71.596	71.007	84.780	63.025
60.000	71.413	71.034	84.986	63.227
61.000	71.212	71.008	86.203	64.440
62.000	71.367	70.950	87.921	66.155
63.000	71.558	70.879	88.231	66.458
64.000	71.488	70.784	86.569	64.762
65.000	71.754	70.709	85.258	63.015
66.000	71.441	70.634	84.084	61.529
67.000	71.560	70.558	82.566	59.863
68.000	71.497	70.520	82.471	59.706
69.000	71.495	70.459	83.413	61.041
70.000	71.495	70.411	83.902	61.879
71.000	71.364	70.446	84.458	62.686
72.000	71.218	70.613	85.264	63.499

• This table shows how prediction error changes from the first forecast step to the last forecast step.
• It is especially useful for explaining whether a model degrades smoothly or sharply as it predicts further into the future.
• Lower is better. This is a percentage-style error that is easier to compare across scales.